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ON THE VALUATIONS CENTERED IN A LOCAL DOMAIN.*¹

By SHREERAM ABHYANKAR.

It is well known that if v is a real zero dimensional valuation of an algebraic function field K/k of n variables and if r is the rational rank of v then the following is true: $r \leq n$ and if $r = n$ then the value group of v is a direct sum of n cyclic groups; see Theorem 1 of [8]. In Section 1, we prove a generalization of this theorem to arbitrary valuations (real or not) centered in an abstract local domain. Namely, let R be a local domain of dimension n with maximal ideal M and quotient field K . Let v be an arbitrary valuation of K having center M in R . Let ρ be the rank and r the rational rank of v , and let d be the transcendence degree of the residue field D of v over R/M . Then we have the following: (1) $d + r \leq n$, (2) if $d + r = n$ then the value group of v is a direct sum of r cyclic groups and $D/(R/M)$ is finitely generated, (3) if $d + \rho = n$ then v is discrete and $D/(R/M)$ is finitely generated.

In Section 2, we generalize some theorems concerning quadratic sequences of quotient rings on a nonsingular algebraic surface to abstract regular local domains. In Theorem 2, we prove that if f is a nonzero element in a two dimensional regular local domain (R, M) with quotient field K and if v is a valuation of K having center M in R then there exists a quadratic transform (R^*, M^*) of R along v and a basis (x^*, y^*) of M^* such that $f = x^{*a}y^{*b}d$ where a and b are nonnegative integers and d is a unit in R^* , (if v is nonreal, then we assume that R is either algebraic or absolute; for definitions see the beginning of Section 1). The special case of Theorem 2 when R is the quotient ring of a point on an algebraic surface and when v is real plays an important role in the proof of the local uniformization theorem (see Proposition 3 of [2] and Lemma 11.2 of [11]). In Theorem 3 we generalize Zariski's factorization theorem on birational transformations between algebraic surfaces to abstract regular two dimensional local domains. As an incidental remark, we show in Proposition 3 that if (R, M) is a regular n dimensional local domain and if w is a valuation of the quotient field of R with center M in R and of R -dimension $n - 1$ then R_w/M_w is a purely transcendental extension.

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sion of a finitely generated extension of R/M . It is a consequence of this proposition that if w is a prime divisor of the second kind having center at a simple subvariety of an algebraic variety then the residue field of w is the function field of a ruled variety. We hope to use the results of this paper in the problem of local uniformization on *absolute* surfaces which we are planning to study.

1. **Classification of valuations.** We start with some remarks about ordered abelian groups. Let G be an ordered abelian group. Let us recall that the rational rank of G is defined to be the maximum cardinal number (finite or infinite) of a subset H of G such that any finite number $h_1, h_2, \dots, h_n$ of distinct elements of H are rationally independent, i.e., if $m_1 h_1 + m_2 h_2 + \dots + m_n h_n = 0$ where the m_i are integers then $m_1 = m_2 = \dots = m_n = 0$. We observe that if G has finite rational rank r then we can find r elements $g_1, g_2, \dots, g_r$ in G such that for any element g in G there exist integers $m_1, m_2, \dots, m_r$ and a nonzero integer m for which $mg = m_1 g_1 + m_2 g_2 + \dots + m_r g_r$ (we call $\{g_1, g_2, \dots, g_r\}$ a rational basis of G); and conversely if there exist r elements $g_1, g_2, \dots, g_r$ in G with this property then G is of finite rational rank $n \leq r$. We assert that the following two conditions are equivalent: (1) G is of finite rational rank r . (2) G is of finite rank ρ and if $(0) = G_0 < G_1 < G_2 < \dots < G_\rho = G$ is the sequence of isolated subgroups of G then G_i/G_{i-1} is of finite rational rank r_i for $i=1, 2, \dots, \rho$. Furthermore, when one and hence both of the above conditions are satisfied, we have the equation: $r = r_1 + r_2 + \dots + r_\rho$. This is easily verified by writing the elements of G as lexicographically ordered ρ -tuples of real numbers² (roughly speaking, if we put together rational bases of $G_1/G_0, G_2/G_1, \dots, G_\rho/G_{\rho-1}$ then we get a rational basis of G). Given an additive group F of real numbers, we shall say that F is an *integral direct sum*, if F is of finite rational rank a and if we can find elements $f_1, f_2, \dots, f_a$ in F such that any element f in F can be expressed as: $f = m_1 f_1 + m_2 f_2 + \dots + m_a f_a$ where the m_i are integers. Again, given an ordered abelian group G , we shall say that G is an *integral direct sum* if G is of finite rational rank r and if G_i/G_{i-1} is an integral direct sum for $i=1, 2, \dots, \rho$ where $(0) = G_0 < G_1 < G_2 < \dots < G_\rho = G$ is the sequence of isolated subgroups of G . One can easily show that G is an integral direct sum if and only if G is of finite rational rank r and if G contains elements $g_1, g_2, \dots, g_r$ such that any element g of G can be expressed as: $g = m_1 g_1 + m_2 g_2 + \dots + m_r g_r$ where the m_i

² See the Appendix.

are integers ($g_1, g_2, \dots, g_r$ are then necessarily rationally independent; we call $\{g_1, g_2, \dots, g_r\}$ an integral basis of G). We observe that G is discrete if and only if G is an integral direct sum and the rational rank of G equals the rank of G . Finally, note that if H is a subgroup of G of finite index then the rank (respectively, the rational rank) of H equals the rank (respectively, the rational rank) of G and H is an integral direct sum (respectively, discrete) if and only if G is an integral direct sum (respectively, discrete).

We shall consistently use, in this paper, the following notations. Let v be a valuation of a field K . We shall denote by R_v the valuation ring of v and by M_v the maximal ideal in R_v . By the rank (respectively, rational rank) of v we shall mean the rank (respectively, rational rank) of the value group of v , also we shall say that v is an integral direct sum if the value group of v is an integral direct sum, and so on. Unless otherwise stated, we shall exclude trivial valuations. When we do want to talk of a trivial valuation, we shall assign to it the zero rank and the zero rational rank, since its value group consists of the zero element alone, and for a trivial valuation we shall consider the designations "integral direct sum" and "discrete" as trivially valid. If v^* is a valuation of a field K^* and v its restriction to a subfield K , then by the v -dimension of v^* is meant the transcendence degree of R_{v^*}/M_{v^*} over R_v/M_v ; if v is trivial (over K) then we have that the v -dimension of v^* equals the K -dimension of v , i.e., the transcendence degree of R_{v^*}/M_{v^*} over K . If R is a local ring and M its maximal ideal, we shall indicate this by saying that (R, M) is a local ring. If a local ring R is the quotient ring of an irreducible subvariety of an algebraic variety, then we shall say that R is algebraic. By Q and Z we shall denote the field of rational numbers and the domain of ordinary integers respectively. If a local ring R is the quotient ring of a domain A finitely generated over Z (i.e., $A = Z[x_1, x_2, \dots, x_n]$ with x_i in A) with respect to a prime ideal in A , then we shall say that R is absolute. If (R, M) is a local domain and v a valuation of the quotient field of R having center in R (i.e., such that $R_v \supset R$ and $R \cap M_v = M$), then by the R -dimension of v is meant the transcendence degree of R_v/M_v over R/M . Recall that if R is a (commutative) ring and P a nonzero prime ideal in R , then by the dimension of P is meant the maximum value of n such that there exists a strictly ascending chain $P = P_0 < P_1 < P_2 < \dots < P_n < R$ of prime ideals P_i , and by the rank of P is meant the maximum value of n such that there exists a strictly descending chain $P = P_0 > P_1 > P_2 > \dots > P_n \supset (0)$ of prime ideals P_i (where P_n is allowed to be the zero ideal if this be prime). Finally, a given integral domain F will be called normal if F is integrally closed in its quotient field.

LEMMA 1. Let K be a field and K^* an extension of K of finite transcendence degree s . Let v^* be a valuation of K^* and let v be the K -restriction of v^* where we allow v to be trivial. Let d be the v -dimension of v^* . Let r and r^* be the rational ranks of v and v^* and let ρ and ρ^* be the ranks of v and v^* respectively. Then we have the following: (1) If r^* is finite, then $r^* + d \leq r + s$. (2) If v is an integral direct sum, if K^*/K is finitely generated, and if $r^* + d = r + s$, then v^* is an integral direct sum and R_{v^*}/M_{v^*} is finitely generated over R_v/M_v (note that $R_v/M_v = K$ if v is trivial). (1*) If ρ is finite, then $\rho^* + d \leq \rho + s$. (3) If v is discrete, if K^*/K is finitely generated, and if $\rho^* + d = \rho + s$, then v^* is discrete and R_{v^*}/M_{v^*} is finitely generated over R_v/M_v .

Proof. First assume that r is finite. We begin by proving the weaker inequality:

$$(A) \quad r^* \leq r + s.$$

Suppose $s = 0$. Given $0 \neq u \in K^*$, let $f(X) = A_0X^n + A_1X^{n-1} + \cdots + A_n$, $A_0 = 1$, be the minimal monic polynomial of u over K . Since $f(u) = 0$, there exist distinct integers i and j such that $v^*(A_iu^{n-i}) = v^*(A_ju^{n-j}) \neq \infty$ and hence $v^*(u) = v(A_i/A_j)/(i-j)$, i.e., the value of u depends rationally on the value of $(A_i/A_j) \in K$. Therefore $r^* = r = r + s$. Now suppose $s > 0$ and assume that (A) is true for $s-1$. Let $z_1, z_2, \dots, z_{s-1}$ be part of a transcendence basis of K^*/K . Let $K_1 = K(z_1, z_2, \dots, z_{s-1})$, let v_i be the restriction of v^* to K_1 (v_i may be trivial), and let r_1 be the rational rank of v_i . By our induction hypothesis, $r_1 \leq r + s - 1$. If the value of every nonzero element of K^* is rationally dependent on the values of elements of K_1 , then $r^* = r_1 \leq r + s - 1 \leq r + s$, and we are through. Now suppose that there is a nonzero element z in K^* such that $h = v^*(z)$ does not depend rationally on the values of elements of K_1 . Then, by the $s = 0$ case, z is transcendental over K_1 . Let $f(X) = f_0 + f_1X + \cdots + f_nX^n$ and $g(X) = g_0 + g_1X + \cdots + g_nX^n$ be nonzero elements of $K_1[X]$. Let $a_i = v^*(f_i)$ if $f_i \neq 0$ and $b_i = v^*(g_i)$ if $g_i \neq 0$. Since h depends rationally neither on the a_i nor on the b_i , there exist integers p and q such that $\infty \neq v^*(f_pz^p) < v^*(f_iz^i)$ whenever $i \neq p$ and $f_i \neq 0$, and $\infty \neq v^*(g_qz^q) < v^*(g_iz^i)$ whenever $i \neq q$ and $g_i \neq 0$; i.e., $v^*(f(z)/g(z)) = v^*(f_p/g_q) + (p-q)h$. Thus, the value of any nonzero element of $K_1(z)$ is of the form $a + mh$ where a is in the value group of v_i and m is an integer, i.e., if r_2 is the rational rank of the restriction of v^* to $K_1(z)$ (this restriction may be trivial), then

$$r_2 = r_1 + 1 \leq r + (s-1) + 1 = r + s.$$

Since $K^*/K_1(z)$ is an algebraic extension, by the case $s=0$, we have $r^*=r_2 \leq r+s$. Thus the induction is complete and (A) has been proved. Also observe that if v_2 is the restriction of v^* to $K_2=K_1(z)$, then the residue fields of v_1 and v_2 coincide. For, in the above notation, since $v_2(f_i z^i) > v_2(f_p z^p)$ whenever $i \neq p$ and $f_i \neq 0$, we must have that $f(z)/(f_p z^p)$ belongs to R_{v_2} and that

$$f(z)/(f_p z^p) = 1 + \sum_{i \neq p} (f_i/f_p) z^{i-p} \equiv 1 \pmod{M_{v_2}}.$$

Similarly, $g(z)/(g_q z^q)$ belongs to R_{v_2} and $g(z)/(g_q z^q) \equiv 1 \pmod{M_{v_2}}$. Now assume that $f(z)/g(z)$ belongs to R_{v_2} . We want to show that we can find e in R_{v_1} with $f(z)/g(z) \equiv e \pmod{M_{v_2}}$. If $f(z)/g(z)$ belongs to M_{v_2} , we can take $e=0$. Now suppose that $f(z)/g(z)$ does not belong to M_{v_2} , i.e., that $v_2(f(z)/g(z)) = 0$. Since $v_2(f(z)/g(z)) = v_2(f_p/g_q) + (p-q)h$ and since h does not depend rationally on $v_2(f_p/g_q)$, we must have $p-q=0$, i.e., that $p=q$ and $v_2(f_p/g_p) = 0$. Let $e = f_p/g_p$. Then

$$f(z)/g(z) = (f_p/g_p) (f(z)/f_p z^p) (g_p z^p/g(z)) \equiv e \pmod{M_{v_2}}$$

since $f(z)/f_p z^p$ and $g_p z^p/g(z)$ are both congruent to one modulo M_{v_2} . This proves our second italicized assertion.

To prove (1) let us retain our assumption that r is finite, and let $\bar{y}_1, \bar{y}_2, \dots, \bar{y}_d$ be a transcendence basis of R_v/M_v over R_v/M_v and fix y_i in R_{v^*} belonging to the residue class $\bar{y}_i$. Let $K' = K(y_1, y_2, \dots, y_d)$ and let v' be the restriction of v^* to K' . Given $0 \neq f(X_1, X_2, \dots, X_d)$ in $K[X_1, X_2, \dots, X_d]$ choose a coefficient q of f having minimum v -value and let $F(X_1, X_2, \dots, X_d) = (1/q) f(X_1, X_2, \dots, X_d)$. Then all the coefficients of $F(X_1, X_2, \dots, X_d)$ belong to R_v and at least one of them is equal to 1. Let $\bar{F}(X_1, X_2, \dots, X_d)$ be the polynomial gotten by reducing the coefficients of $F(X_1, X_2, \dots, X_d)$ modulo M_v . Since $\bar{F}(X_1, X_2, \dots, X_d)$ has a coefficient equal to 1 and since $\bar{y}_1, \bar{y}_2, \dots, \bar{y}_d$ are algebraically independent over R_v/M_v , we must have $\bar{F}(\bar{y}_1, \bar{y}_2, \dots, \bar{y}_d) \neq 0$, i.e., $v^*(F(y_1, y_2, \dots, y_d)) = 0$, i.e., $v^*(f(y_1, y_2, \dots, y_d)) = v(q) \neq \infty$, and hence $f(y_1, y_2, \dots, y_d) \neq 0$. Thus $y_1, y_2, \dots, y_d$ are algebraically independent over K and the value groups of v and v' are identical. Since the transcendence degree of K^*/K' is $s-d$, (1) follows by applying (A) to K^*/K' . Now let $g(X_1, X_2, \dots, X_d)$ and $h(X_1, X_2, \dots, X_d)$ be arbitrary nonzero elements of $K[X_1, X_2, \dots, X_d]$ and let

$$y = f(y_1, y_2, \dots, y_d)/g(y_1, y_2, \dots, y_d).$$

Fix coefficients a and b of g and h respectively having minimum v' -values

and let $p = a/b$. Let $G(X_1, X_2, \dots, X_d) = (1/a)g(X_1, X_2, \dots, X_d)$ and $H(X_1, X_2, \dots, X_d) = (1/b)h(X_1, X_2, \dots, X_d)$. Then, as above,

$$v'(g(y_1, y_2, \dots, y_d)/h(y_1, y_2, \dots, y_d)) = v'(a/b).$$

Hence $y = g(y_1, y_2, \dots, y_d)/h(y_1, y_2, \dots, y_d)$ belongs to R_v if and only if $p = a/b$ belongs to $R_v \cap K = R_v$. Now assume that y does belong to R_v . Let $\bar{y}$ and $\bar{p}$ be the residue classes modulo M_v containing y and p respectively. Let $\bar{G}(X_1, X_2, \dots, X_d)$ and $\bar{H}(X_1, X_2, \dots, X_d)$ be the polynomials obtained respectively from $G(X_1, X_2, \dots, X_d)$ and $H(X_1, X_2, \dots, X_d)$ by reducing their coefficients modulo M_v . Since $\bar{H}$ has a coefficient equal to one and since $\bar{y}_1, \bar{y}_2, \dots, \bar{y}_d$ are algebraically independent over R_v/M_v , we have that

$$\bar{y} = \bar{p}\bar{G}(\bar{y}_1, \bar{y}_2, \dots, \bar{y}_d)/\bar{H}(\bar{y}_1, \bar{y}_2, \dots, \bar{y}_d).$$

Therefore $R_v/M_v = (R_v/M_v)(\bar{y}_1, \bar{y}_2, \dots, \bar{y}_d)$, and hence in particular R_v/M_v is finitely generated over R_v/M_v .

Now assume that v is an integral direct sum, that K^*/K is finitely generated, and that $r^* + d = r + s$. Let K' and v' be as above. Then v and v' have the same value groups, K^*/K' is a finitely generated extension of transcendence degree $e = s - d = r^* - r$, and R_v/M_v is finitely generated over R_v/M_v . Fix an integral basis $t_1, t_2, \dots, t_r$ of the value group of v' . Let $x_1, x_2, \dots, x_e$ be a transcendence basis of K^*/K' . Let $K'_i = K'(x_1, x_2, \dots, x_i)$, $v'_i =$ the restriction of v^* to K'_i , and $r'_i =$ the rational rank of v'_i . Since $r^* = r + e$, we must have, in view of (1), $r_i = r_{i-1} + 1$ for $i = 1, 2, \dots, e$. Let $v^*(x_i) = t_{r+1}$. By applying the first of the above italicized remarks successively to the extensions $K'_1/K', K'_2/K'_1, \dots, K'_e/K'_{e-1}$, we conclude that for any nonzero element x of K'_e we have

$$v^*(x) = a + m_{r+1}t_{r+1} + m_{r+2}t_{r+2} + \dots + m_{r+e}t_{r+e},$$

where a is the value of an element of K' and where $m_{r+1}, m_{r+2}, \dots, m_{r+e}$ are integers; since $a = m_1t_1 + m_2t_2 + \dots + m_rt_r$ where $m_1, m_2, \dots, m_r$ are integers, we finally have: $v^*(x) = m_1t_1 + m_2t_2 + \dots + m_{r+e}t_{r+e}$. Therefore v_e^* is an integral direct sum. Since K^*/K'_e is a finite algebraic extension, the value group of v_e^* is a subgroup of the value group of v^* of finite index and hence v^* is an integral direct sum. Now by the second italicized remark above, the residue field of v_e^* coincides with the residue field of v' . Since the residue field of v' is finitely generated over the residue field of v and since K^*/K'_e is a finite algebraic extension, we conclude that R_v/M_v is finitely generated over R_v/M_v . This proves (2).

The proof of (1*) is entirely similar to that of (1). Finally, assume

that v is discrete, $\rho^* + d = \rho + s$, and that K^*/K is finitely generated. The discreteness of v implies that $\rho = r$. Since by (1), $r^* + d \leq r + s$ and since $r^* \geq \rho^*$, it follows that $r^* = \rho^*$ and that $r^* + d = r + s$. Hence by (2), v^* is an integral direct sum and R_{v^*}/M_{v^*} is finitely generated over R_v/M_v . Since $r^* = \rho^*$, v^* is discrete. This proves (3).

COROLLARY 1. *In the notation of the above lemma, assume that v is trivial, i. e., that v^* is a valuation of K^*/K . Then: (1) $\rho^* + d \leq r^* + d \leq s$. Furthermore, if K^*/K is finitely generated (i. e., if K^*/K is an algebraic function field of dimension s), then we have the following: (2) If $r^* + d = s$ then v^* is an integral direct sum and R_{v^*}/M_{v^*} is finitely generated over K . (3) if $\rho^* + d = s$ then v^* is discrete and R_{v^*}/M_{v^*} is finitely generated over K . (4) If $d = s - 1$ then v^* is real discrete and R_{v^*}/M_{v^*} is finitely generated over K .*

PROPOSITION 1. *Let K be a field which is of finite transcendence degree over its prime field P . Let n be the absolute dimension³ of K and let v be a valuation of K . Let r be the rational rank of v , let ρ be the rank of v , and let d be the absolute dimension of R_v/M_v . Then: (1) $r + d \leq n$. Furthermore, if K is finitely generated over P then we have the following: (2) If $r + d = n$ then v is an integral direct sum and R_v/M_v is finitely generated over its prime field. (3) If $\rho + d = n$ then v is discrete and R_v/M_v is finitely generated over its prime field. (4) If $d = n - 1$ then v is real discrete and R_v/M_v is finitely generated over its prime field.*

Proof. Let w be the restriction of v to P where w may be trivial. If P is of nonzero characteristic then w is trivial, $n =$ the transcendence degree of K over P , and $R_w/M_w = P$; and hence the proposition follows from the previous corollary. Now assume that P is of characteristic zero, i. e., that $P = Q$ (the field of rational numbers). Let D be the w -dimension of v , d^* the absolute dimension of R_w/M_w , and r^* the rational rank of w . If w is trivial, then $r^* = 0$ and $R_w/M_w = Q$, i. e., $d^* = 1$ and hence $r^* + d^* = 1$. If w is nontrivial, then w must be the real discrete valuation given by a prime number p of Z and hence $r^* = 1$ and $R_w/M_w = Z/(pZ)$, i. e., $d^* = 0$ and hence $r^* + d^* = 1$. Thus in both the cases, w is discrete, $r^* + d^* = 1$, and R_w/M_w is its own prime field. Hence $r + D \leq r^* + (n - 1)$ if and only if $r + (d - d^*) \leq r^* + (n - 1)$, i. e., if and only if

$$r + d \leq (d^* + r^*) + (n - 1) = 1 + (n - 1) = n.$$

³ Let D be an integral domain with quotient field K such that K is of finite transcendence degree s over its prime subfield. Then we define the absolute dimension of D to be s or $s + 1$ according as K is of nonzero or zero characteristic respectively.

Thus $r + D \leq r^* + (n-1)$ if and only if $r + d \leq n$ and $r + D = r^* + (n-1)$ if and only if $r + d = n$. Hence (1), (2), (3) follow respectively from parts (1), (2), (3) of Lemma 1 applied to the extension K/Q . Finally, (4) follows from (1) and (3).

DEFINITION 1. In the notation of Proposition 1, if $d = n-1$ then we shall say that v is a prime divisor of K .

Remark 1. Let K be a field of absolute dimension two which is finitely generated over its prime field P of characteristic p ; we can express this by saying that K is the function field of an absolute surface of characteristic p . Let v be a valuation of K . It follows by Proposition 1 that v is of one of the following four types: (1) v is a prime divisor of K , i.e., v is real discrete and R_v/M_v is of absolute dimension one. If $p \neq 0$ then R_v/M_v is an algebraic function field of one variable over P . If $p = 0$ then R_v/M_v is either a finite algebraic extension of Q or it is an algebraic function field of dimension one over a prime field of nonzero characteristic. In either case, R_v/M_v is the function field of an absolute curve. (2) v is discrete of rank two. (3) v is real of rational rank one, but v is not a divisor of K . (4) v is real of rational rank two; in this case v is necessarily an integral direct sum. In cases (2), (3), and (4), R_v/M_v is an algebraic extension of a prime field of characteristic $q \neq 0$ and hence R_v/M_v is perfect; if $p \neq 0$ then $q = p$; in cases (2) and (4), R_v/M_v , being finitely generated over its prime field, is necessarily finite.

If $p \neq 0$ then K/P is an algebraic function field of two variables, i.e., an absolute surface (respectively, an absolute n -dimensional variety) of characteristic $p \neq 0$ is simply an algebraic surface (respectively, an algebraic n -dimensional variety) over the prime field of characteristic p ; in this case the above classification of valuations is well known. The present remark signifies that a parallel situation prevails for absolute surfaces of characteristic zero, i.e., algebraic curves over Q considered as surfaces.

PROPOSITION 2. Let (R, M) be a local domain of dimension n with quotient field K . Let v be a real valuation of K with center M in R . Let d be the R -dimension of v and let r be the rational rank of v . Then $d + r \leq n$. Furthermore, if $d + r = n$ then v is an integral direct sum and R_v/M_v is finitely generated over R/M .

We shall divide the proof of this proposition into several lemmas.

LEMMA 2. Proposition 2 is true in case R is the power series ring $k[[x_1, x_2, \dots, x_n]]$ in n variables $x_1, x_2, \dots, x_n$ over a field k .

Proof. Let

$$K^* = k(x_1, x_2, \dots, x_n), \quad A = k[x_1, x_2, \dots, x_n], \quad P = (x_1, x_2, \dots, x_n)A,$$

$R^* = A_P$, $M^* = PR^*$, and v^* = the restriction of v to K^* . Then (R, M) is the completion of (R^*, M^*) and hence by Lemma 12 of [2], v^* has center M^* in R^* and v^* has the same residue field and value group as v . Since $R/M = R^*/M^*$, d is also the v^* -dimension of v^* . Since $R^*/M^* = k$, the present lemma follows by applying Lemma 1 to the extension K^*/k .

LEMMA 3. *Proposition 2 is true in case R is the power series ring $R_w[[x_1, x_2, \dots, x_{n-1}]]$ in $n-1$ variables $x_1, x_2, \dots, x_{n-1}$ over a complete valuation ring R_w of a real discrete valuation w such that R_w is of characteristic zero and $M_w = pR_w$ where p is an ordinary prime number.*

Proof. Let k be the quotient field of R_w , $K^* = k(x_1, x_2, \dots, x_{n-1})$, $A = R_w[x_1, x_2, \dots, x_{n-1}]$, $P = A \cap M$, $R^* = A_P$, $M^* = PR^*$, and v^* = the K^* -restriction of v . Then it is easily verified that $M^* = (x_1, x_2, \dots, x_{n-1})R^*$, that (R^*, M^*) is a regular local domain of dimension n , and that (R, M) is the completion of (R^*, M^*) ; [Proof: $M = (x_1, x_2, \dots, x_{n-1}, p)R$, $M \cap R^* = M^*$ and hence R and R^* are concordant, and so on]. By the argument used in the proof of Lemma 12 of [2], it follows that v^* has center M^* in R^* and v^* has the same value group and residue field as v . Since $R/M = R^*/M^*$, d must also be the R^* -dimension of v^* . Since $M_v \cap R_w = M^* \cap R_w = M_w$ and $R^*/M^* = R_w/M_w$ (Lemma 5 of [3]), since $R_w \subset R_{v^*}$ and since R_w is a maximal subring of k , it follows that w is the k -restriction of v^* and that d is also the w -dimension of v^* . The present lemma now follows by applying Lemma 1 to the extension K^*/k .

LEMMA 4. *Proposition 2 is true if R is an unramified complete regular local ring.*

Proof. By Theorem 15 of [3], R is isomorphic either to the ring described in Lemma 2 or to the ring described in Lemma 3. Therefore, Lemma 4 follows from Lemmas 2 and 3.

LEMMA 5. *Proposition 2 is true if R is complete.*

Proof. By Theorem 16 of [3], R is a finite module over a subring S which is an unramified complete regular local ring having the same dimension and residue field as R . Hence Lemma 5 follows from Lemma 4 by observing that valuations do not change their character in passing to a finite algebraic extension.

LEMMA 6. *Proposition 2 is true.*

Proof. Since v is real, we can employ the technique used in the proof of Theorem 1 of [13] to pass from the case of complete local domains to the general case; Lemma 6 then follows from Lemma 5.

THEOREM 1. *Let (R, M) be a local domain of dimension n with quotient field K and let v be a valuation of K with center M in R . Let d , r , and ρ be respectively the R -dimension, the rational rank, and the rank of v . Then:*
 (1) $d + r \leq n$. (2) *If $d + r = n$ then v is an integral direct sum and R_v/M_v is finitely generated over R/M . (3) If $d + \rho = n$ then v is discrete and R_v/M_v is finitely generated over R/M . (4) If $d = n - 1$ then v is real discrete and R_v/M_v is finitely generated over R/M .*

Proof. We observe that (3) and (4) follow at once from (1) and (2), and we proceed to prove (1) and (2) by applying induction to n . Suppose first that $n = 1$. Let D be the integral closure of R in K . Then D is a Dedekind domain (§ 39 of [5]) and hence we must have $R_v \supset D$ and $M_v \cap D =$ a minimal prime ideal H in D , and hence $D_H \subset R_v$. Since D_H is a real discrete valuation ring, it is a maximal subring of K and hence $D_H = R_v$, i.e., v is real discrete. Hence, for $n = 1$, (1) and (2) follow from Proposition 2. Now let $n > 1$ and assume that (1) and (2) are true for all smaller values of n . If v is real then (1) and (2) follow from Proposition 2. So assume that v is nonreal. Given a nonzero prime ideal E in R_v , $(R_v)_E$ is the valuation ring of a valuation of K (see [4]), and hence $E \cap R$ must be a nonzero prime ideal in R (since K is the quotient field of R). Since the set T of nonzero prime ideals in R is simply ordered by inclusion [4], the set T^* of the R -contractions of the members of T is also simply ordered by inclusion. Since any chain of prime ideal in R is of length at most n , T^* must be a finite set. Therefore $P = \bigcap_{F \in T^*} F$ is the smallest ideal in T^* , and hence P is a nonzero prime ideal. Let $B = \bigcap_{E \in T} E$. Then $B \cap R = P \neq (0)$ and hence $B \neq (0)$. Therefore $(R_v)_B$ is the valuation ring of a nontrivial valuation w of K . Since B is the minimal nonzero prime ideal of R_v , w must be real. Let $K^* = R_w/M_w$, $\bar{R} = R/P$, $\bar{M} = M/P$, $\bar{K}$ = the quotient field of $\bar{R}$ in K^* , v^* = the valuation of K^* induced by v (v^* is nontrivial), $\bar{v}$ = the restriction of v^* to $\bar{K}$ where $\bar{v}$ may be trivial. Let $S = R_P$ and $N = PS$. Then w has center N in S . Let d' be the S -dimension of w . Then d' = the transcendence degree of $K^*/\bar{K}$, since $\bar{K} = S/N$. Let $\bar{r}$, r^* , and r' be the rational ranks of $\bar{v}$, v^* , and w respectively. We observe that $r = r^* + r'$

and we divide the remaining argument into two cases according as $P = M$ or $P \neq M$.

Case 1, $P = M$. Then $\bar{R} = \bar{K}$, $\bar{M} = (0)$ and $\bar{v}$ is trivial. Also, the $\bar{K}$ -dimension of v^* = the $\bar{R}$ -dimension of $v = d$. Applying Proposition 2 to the real valuation w we get $d' + r' \leq n$. Applying Corollary 1 of Lemma 1 to the extension $K^*/\bar{K}$ we get $r^* + d \leq d'$. Therefore $r + d = r^* + r' + d \leq n$. Now assume that $r + d = n$. Then $d' + r' = n$ and $r^* + d = d'$. Since $d' + r' = n$, it follows by Proposition 2 that w is an integral direct sum and $K^*/\bar{K}$ is finitely generated. Since $r^* + d = d'$ and since $K^*/\bar{K}$ is finitely generated, it follows by Corollary 1 of Lemma 1 that v^* is an integral direct sum and $R_{v^*}/M_{v^*} = R_v/M_v$ is finitely generated over $\bar{K} = R/M$. Since v is composed of v^* and w and since v^* and w are integral direct sums, it follows that v is also an integral direct sum.

Case 2, $P \neq M$. Let $\bar{n}$ and m be the dimensions of the local rings $\bar{R}$ and S respectively. Then $\bar{n} = \dim P$ and $m = \text{rank } P$. Therefore

$$(A) \quad \bar{n} + m \leq n.$$

Since $m > 0$, we must have $0 < \bar{n} < n$. Now $\bar{v}$ is a nontrivial valuation of $\bar{K}$ having center $\bar{M}$ in $\bar{R}$. Let $\bar{r}$ be the rational rank of $\bar{v}$ and let $\bar{d}$ be the $\bar{R}$ -dimension of $\bar{v}$. Let d^* be the $\bar{v}$ -dimension of v^* . Then

$$(B) \quad \bar{d} + d^* = d.$$

Since w is a real valuation with center N in S , it follows by Proposition 2 that

$$(C) \quad r' + d' \leq m.$$

Applying our induction hypothesis to the valuation $\bar{v}$ having center $\bar{M}$ in $\bar{R}$ we get

$$(D) \quad \bar{r} + \bar{d} \leq \bar{n}.$$

By Lemma 1,

$$(E) \quad r^* + d^* \leq \bar{r} + \bar{d}.$$

Adding the inequalities (C), (D), and (E) we get

$$r' + \bar{r} + r^* + d' + \bar{d} + d^* \leq m + \bar{n} + \bar{r} + d',$$

i. e.,

$$r' + r^* + \bar{d} + d^* \leq m + \bar{n}.$$

Since $r' + r^* = r$ and since by (B), $\bar{d} + d^* = d$, we obtain:

$$(F) \quad r + d \leq m + \bar{n}.$$

Therefore, in view of (A), we conclude that $r + d \leq n$. Now assume that $r + d = n$. Then by (A) and (F) we must have $r + d = m + \bar{n}$. Since $r = r' + r^*$ and since by (B), $d = \bar{d} + d^*$, we obtain:

$$r' + \bar{r} + r^* + d' + \bar{d} + d^* = m + \bar{n} + \bar{r} + d'.$$

Therefore by (C), (D), and (E), it follows that:

$$(C') \quad r' + d' = m.$$

$$(D') \quad \bar{r} + \bar{d} = \bar{n}.$$

$$(E') \quad r^* + d^* = \bar{r} + d'.$$

By our induction hypothesis, (D') implies that $\bar{v}$ is an integral direct sum and that $R_{\bar{v}}/M_{\bar{v}}$ is finitely generated over $\bar{R}/\bar{M} = R/M$. Therefore, in view of (E'), it follows by Lemma 1, that v^* is an integral direct sum and that $R_v/M_v = R_{v^*}/M_{v^*}$ is finitely generated over $R_{\bar{v}}/M_{\bar{v}}$ and hence over R/M . Again, in view of (C'), it follows by Proposition 2 that w is an integral direct sum. Since w and v^* are both integral direct sums, we finally conclude that v is also an integral direct sum.

Thus the induction is complete and the theorem has been established.

DEFINITION 2. In the notation of Theorem 1, if v is of R -dimension $n-1$ then we shall say that v is a prime divisor for (R, M) .

Remark 2. In the notation of Theorem 1, let $n=2$. Then we get a classification of valuations v of K centered in R which is parallel to the one given in Remark 1 for the case of absolute surfaces. Namely, v is of one of the following four types: (1) v is a prime divisor for (R, M) , i. e., $d=1$ and v is real discrete. (2) $d=0$ and $\rho=2$; in this case, necessarily, $r=\rho=2$, i. e., v is discrete. (3) $d=0$ and $\rho=r=1$. (4) $d=0$, $\rho=1$ and $r=2$; in this case v is necessarily an integral direct sum. In cases (1), (2), and (4), R_v/M_v is necessarily finitely generated over R/M . Also note that if v is of positive R -dimension, then v is necessarily of type (1), i. e., v is a prime divisor.

2. Quadratic transformations.

LEMMA 7. Let $R_1 \subset R_2 \subset \dots$ be a sequence of normal integral domains with a common quotient field K such that R_i contains a unique maximal ideal M_i and $M_{i+1} \cap R_i = M_i$ for $i=1, 2, \dots$. Assume that $\bigcup_{i=1}^{\infty} R_i$ is not the valuation ring of a valuation of K . Then there exist infinitely many valua-

tions w of K which have center M_i in R_i and for which R_w/M_w is of positive transcendence degree over R_i/M_i for each i .

Proof. The proof is essentially the same as the one given by Zariski in the special case when the (R_i, M_i) are quotient rings of corresponding subvarieties under a sequence of birational transformations, see the theorem on page 25 of [14]. To outline the main idea of the proof, first observe that for an arbitrary commutative ring A with identity the following two statements are equivalent: (1) A contains a unique maximal ideal. (2) The set of nonunits in A is an ideal.

Now let $R = \bigcup_{i=1}^{\infty} R_i$ and $M = \bigcup_{i=1}^{\infty} M_i$. We may canonically assume that $R_1/M_1 \subset R_2/M_2 \subset \cdots$. Let $D = \bigcup_{i=1}^{\infty} R_i/M_i$. Then D is a field and $R/M = D$, i.e., M is a maximal ideal in the domain R . Also observe that R is normal. Since R is not a valuation ring, there exists $x \in K$ with $x \notin R$ and $(1/x) \notin R$. The canonical homomorphism h of R onto D can be uniquely extended to a homomorphism H of $R[x]$ onto $D[x]$ for which $H(x) = X$, where X is a transcendental over D ; see pp. 26-27 of [14]. Let p be any one of the infinitely many prime ideals in $D[X]$. Let $P = H^{-1}(p)$. By the theorem of existence of valuations, there exists a valuation w of K having center P in $R[x]$. Since $R_w/M_w \supset D(x)$, it is clear that w has the required properties. The infinitely many choices of p gives us infinitely many w of the required type.

LEMMA 8. Let (R, M) be a two dimensional normal local domain with quotient field K . Let P be a minimal prime ideal in R . Then: (1) R_P is the valuation ring of a real discrete valuation w of K ; (2) there exists at least one and at most a finite number of valuations v of K having center M in R which are composed with w , i.e., for which $R_v \subset R_w$; and (3) each such valuation v is discrete of rank two and R_v/M_v is a finite algebraic extension of R/M (hence in particular v is of R -dimension zero).

Proof. (1) is proved on page 103 of [5]. The proof of (2) is the same as in Lemma 11 of [2] and is as follows: $R/(R \cap M_w)$ is a local domain of dimension one with quotient field R_w/M_w . Hence the integral closure of $R/(R \cap M_w)$ in R_w/M_w is a Dedekind domain D with a finite number of prime ideals $P_1, P_2, \dots, P_h$ (§39 of [5]). Let v_i^* be the real discrete valuation of R_w/M_w with $R_{v_i^*} = D_{P_i}$. Let v_i be the valuation of R_w/M_w which is composed of w and v_i^* . Then $v_1, v_2, \dots, v_h$ are exactly the valuations described in (2). Finally, (3) follows at once from Theorem 1.

LEMMA 9. Let (R, M) be a two dimensional regular local domain with quotient field K . Then: (1) R is a unique factorization domain, i. e., equivalently, every minimal prime ideal in R is principal; and (2) there exist infinitely many valuations of K having center M in R which have R -dimension zero and which are discrete of rank two.

Proof. (1) follows by Satz 8 and Satz 9 of Krull [7].⁴ To prove (2) it is enough to show, in view of Lemma 8, that there exist infinitely many relatively prime irreducible nonunits in R . To show this, let x, y be a minimal basis of M . Let $P = xR$, $\bar{R} = R/P$, $\bar{M} = M/P$, $\bar{K} = R_P/(PR_P)$, $\bar{y}$ = the residue class modulo P containing y , and w = the real discrete valuation of $\bar{K}$ with $R_w = R_P$. Then $\bar{M} = \bar{y}\bar{R}$ and hence $\bar{R}$ is a regular one dimensional local domain, i. e., $\bar{R} = R_v$ where v is a real discrete valuation of $\bar{K}$ with $\bar{v}(\bar{y}) = 1$. Let v be the valuation of K which is composed of w and $\bar{v}$, and let us write the elements of the value group of v as lexicographically ordered pairs of integers. Let $x_m = x + y^m$ where m is a positive integer. Since (x_m, y) is a basis of M , x_m is an irreducible nonunit. Since $v(x_m) = (0, \bar{v}(\bar{y}^m)) = (0, m)$, we have that $v(x_m) \neq v(x_n)$ whenever $m \neq n$. Therefore $x_1, x_2, x_3, \dots$ are infinitely many pairwise relatively prime irreducible nonunits in R .

Now let us recall the notion of a quadratic transform.

LEMMA 10. Let (R, M) be a regular local domain of dimension $s > 1$ and let $x_1, x_2, \dots, x_s$ be a minimal basis of M . Let v be a valuation of the quotient field K of R having center M in R . Suppose we have arranged the x_i so that $v(x_1) \leq v(x_i)$ for $i = 1, 2, \dots, s$. Let $A = R[x_2/x_1, x_3/x_1, \dots, x_s/x_1]$, $P = A \cap M_v$, $S = A_P$ and $N = PS$. Then (S, N) is a regular local domain of dimension $t \leq s$, v has center N in S and if by d and d^* we denote respectively the R -dimension and the S -dimension of v then we have: $s - t = d - d^*$.

Proof. By Lemma 3 of [1], $A/(x_1A)$ can be canonically identified with a polynomial ring $A^* = (R/M)(X_2, X_3, \dots, X_s)$ in $s-1$ variables over R/M . Let $P^* = P/(x_1A)$. Then $A^*_{P^*}$ is a regular local ring of dimension $h \leq s-1$. Fix $y_2^*, y_3^*, \dots, y_{h+1}^*$ in A^* such that

$$P^*A^*_{P^*} = (y_2^*, y_3^*, \dots, y_{h+1}^*)A^*_{P^*}.$$

⁴We take this opportunity in pointing out that there is obviously a misprint in the statement of Satz 9 of Krull [7], namely that the term "Stellenring" should be replaced by "p-reihenring." For in the proof it is used that the Anfang forms are binary which is true only for p-reihenrings; and further, the theorem is certainly not true for arbitrary two dimensional complete local rings.

Fix y_i in A belonging to the residue class y_i^* . Then it is clear that $(x_1, y_2, y_3, \dots, y_{h+1})S = N$ and that

$$t = \dim S = \text{rank } N = 1 + \text{rank } P^* = 1 + h.$$

Therefore S is regular. Also $t \leq 1 + h \leq (s-1) + 1 = s$. Since $N \cap R = M$, we may canonically assume that $R/M \subset S/N \subset R_0/M_0$. Since the transcendence degree of $S/N = A^*/P^*$ over R/M is $s-1-h=s-t$, we conclude that: $s-t = (R\text{-dimension of } v) - (S\text{-dimension of } v) = d-d^*$.

DEFINITION 3. In the notation of the above lemma, S is called "the first (or immediate) quadratic transform of R along v ." Let now $R_0 = R$ and let R_i be the first quadratic transform of R_{i-1} along v assuming that $\dim R_{i-1} > 1$. If $\dim R_i > 1$, for $i=1, 2, \dots, n-1$ then R_n will be defined and we shall say that " R_n is the n -th quadratic transform of R along v ." If S is the n -th quadratic transform of R along v for some n , we shall say that " S is a quadratic transform of R along v ." Finally, if S is a quadratic transform of R along some valuation v having center M in R , we shall say that " S is a quadratic transform of R ."

DEFINITION 4. Let (R, M) and (S, N) be local domains with a common quotient field K . If $S \supset R$ and $N \cap R = M$ then we shall say that (S, N) has center M in R . If (S, N) has center M in R and if there exists a finite set of elements $x_1, x_2, \dots, x_n$ in S such that $S = A_P$ and $N = PS$ where $A = R[x_1, x_2, \dots, x_n]$ and $P = A \cap N$ then we shall say that " (S, N) is a finite transform of (R, M) ."

LEMMA 11. Let (R, M) and (S, N) be local domains with a common quotient field K . Then we have the following: (1) If S is a finite transform of R and if R is absolute (respectively, algebraic) then S is absolute (respectively, algebraic). (2) If R is regular and if S is a quadratic transform of R then S is a finite transform of R .

Proof. Assume that S is a finite transform of R and fix $x_1, x_2, \dots, x_n$ in S such that $S = A_P$ and $N = PS$ where $A = R[x_1, x_2, \dots, x_n]$ and $P = A \cap N$. Let k = the ground field or $k = Z$ according as R is algebraic or absolute respectively. Then we can find a finite number of elements $x_{n+1}, x_{n+2}, \dots, x_m$ in R such that $R = A^*_{P^*}$ where $A^* = k[x_{n+1}, x_{n+2}, \dots, x_m]$ and $P^* = A^* \cap M$. Let $A' = k[x_1, x_2, \dots, x_m]$ and $P' = A' \cap N$. Then it is easily seen that $S = A'_P$ and $N = P'S$ and hence S is absolute or algebraic according as R is absolute or algebraic respectively. The proof of (2) is entirely similar.

PROPOSITION 3. Let (R, M) be an n -dimensional regular local domain with quotient field K with $n > 1$, and let v be a valuation of K which is a prime divisor for R . Then the quadratic sequence along v starting from R is necessarily finite, i. e., if $R = R_0$ and if R_i is the first quadratic transform of R_{i-1} along v provided $\dim R_{i-1} > 1$ then for some integer h we have that R_h is one dimensional; we also have: $R_h = R_v$. Furthermore, there exists a field T with $R/M \subset T \subset R_v/M_v$ such that T is finitely generated over R/M and R_v/M_v is a pure transcendental extension of T of finite positive transcendence degree (we may express this by saying that " R_v/M_v is a ruled extension of R/M ").

Proof. Assume the contrary, i. e., that the quadratic sequence $R = R_0 < R_1 < R_2 < \dots$ along v is infinite. It then follows, by Lemma 10, that there exists an integer s such that $\dim R_t = \dim R_s$ and R_t -dimension of $v = m - 1$ whenever $t \geq s$, where we have set $m = \dim R_s$. Let $S = \bigcup_{i=0}^{\infty} R_i$ and $N = \bigcup_{i=0}^{\infty} M_i$ where M_i is the maximal ideal in R_i . Then, as in the proof of Lemma 7, N is the unique maximal ideal in S and $\bigcup_{i=s}^{\infty} R_i/M_i = S/N$. Since, as in the proof of Lemma 10, R_{t+1}/M_{t+1} is an algebraic extension of R_t/M_t whenever $t \geq s$, it follows that S/N is an algebraic extension of R_t/M_t for any $t \geq s$. Now v has center M_i in R_i for each i and by Theorem 1, v is real discrete. Hence, if S were not the valuation ring of a valuation of K then, following the considerations of pages 27-28 of [14], we would reach a contradiction. Therefore, S must be the valuation ring of a valuation w of K . Since R_s has a first quadratic transform R_{s+1} , it follows by the definition of quadratic transforms that $m > 1$, i. e., that v is of positive R_t -dimension whenever $t \geq s$. Now $R_v \supset R_w$ and

$$R_w \cap M_v = S \cap M_v = \left(\bigcup_{i=0}^{\infty} R_i \right) \cap M_v = \left(\bigcup_{i=0}^{\infty} (R_i \cap M_v) \right) = \bigcup_{i=0}^{\infty} M_i = N = M_w.$$

Therefore $v = w$, i. e., $R_v/M_v = S/N$ is an algebraic extension of R_s/M_s . Thus our assumption that the quadratic sequence $R_0 < R_1 < R_2 < \dots$ is infinite is absurd. Let R_h be the last member of this sequence. Then R_h is a regular one dimensional local domain, i. e., R_h is the valuation ring of a real discrete valuation of K . Since $R_h \subset R_v$, we must have $R_h = R_v$. Now let $T = R_{h-1}/M_{h-1}$ and let d be the dimension of R_{h-1} . Then $d > 1$. Let $x_1, x_2, \dots, x_d$ be a minimal basis of M_{h-1} arranged so that $v(x_1) \leq v(x_i)$ for $i = 1, 2, \dots, d$. Let $A = R_{h-1}[x_2/x_1, x_3/x_1, \dots, x_d/x_1]$ and $P = A \cap M_h$. As in Lemma 3 of [1], we may identify $A/(x_1 A)$ with the polynomial ring

$A^* = T[Y_2, Y_3, \dots, Y_d]$. Let $P^* = P/(x_1A)$. As in the proof of Lemma 10, $\text{rank } P^* = (\dim R_h) - 1 = 0$, i. e., $P^* = (0)$. Since $R_v/M_v = R_h/M_h$ is isomorphic to A^*/P^* , i. e., to $T(Y_2, Y_3, \dots, Y_d)$, we conclude that R_v/M_v is a pure transcendental extension of T of transcendence degree $d-1 > 0$. By Theorem 1, R_v/M_v is finitely generated over R/M and hence T is also finitely generated over R/M ; this also follows from the fact that R_{h-1} is a finite transform of R .

PROPOSITION 4. *Let V be an r -dimensional algebraic variety with function field K/k , let v be a prime divisor of K/k , and let W be the center of v on V . Let s be the dimension of W . Assume that v is of the second kind for V (i. e., that $s < r-1$) and that W is simple for V . Let V^* be any other projective model of K/k such that V is of the first kind for V^* and such that V^* is normal at the center of W^* of v on V^* . Let L/k be the function field of W^* . Then we have the following: (1) W^* is a ruled variety, i. e., we can find a field T with $k \subset T \subset L$ such that L is a pure transcendental extension of T of positive transcendent degree. (2) If $r=2$, then we can find a field T with $k \subset T \subset L$ such that T/k is a finite algebraic extension and L/T is a pure transcendental extension of transcendence degree one. (3) If $r=2$ and if k is algebraically closed, then L/k is a pure transcendental extension of transcendence degree one, i. e., W^* is a rational curve.*

Proof. (1) follows from Proposition 3 by observing that $R_v/M_v = L$ and that if (R, M) denotes the quotient ring of W on V then $k \subset R/M \subset R_v/M_v$. Again, (2) follows from (1) and (3) follows from (2).

LEMMA 12. *Let $R_0 < R_1 < R_2 < \dots$ be a strictly ascending sequence of two dimensional regular local domains with a common quotient field K , let M_i be the maximal ideal in R_i , and let $S = \bigcup_{i=0}^{\infty} R_i$. Assume that R_{i+1} is a quadratic transform of R_i for $i=0, 1, 2, \dots$. Then: (1) S is the valuation ring of a valuation v^* of K such that v^* has center M_i in R_i and such that v^* is of R_i -dimension zero for each i . Furthermore: (2) if v is any valuation of K with center M_i in R_i for $i=0, 1, 2, \dots$, then $v = v^*$.*

Proof. By introducing all the intermediate successive quadratic transforms between R_i and R_{i+1} for each i , we may assume without loss of generality, since this would not change S , that R_{i+1} is a first quadratic transform of R_i for $i=0, 1, 2, \dots$. Now suppose, if possible, that S is not the valuation ring of a valuation of K . Then by Lemma 7, there exists a valuation w of K having center M_i in R_i and of positive R_i -dimension for each i ; and hence

by Theorem 1, w is a prime divisor for R_0 . Therefore by Proposition 3, the sequence $R_0 < R_1 < R_2 < \dots$ must be finite, which is absurd. Hence $S = R_v$, where v^* is a valuation of K . Since, as in the proof of Lemma 7, $M_v = \bigcup_{i=0}^{\infty} M_i$, it follows that

$$R_i \cap M_v = R_i \cap \left(\bigcup_{j=0}^{\infty} M_j \right) = R_i \cap \left(\bigcup_{j=i+1}^{\infty} M_j \right) = \bigcup_{j=i+1}^{\infty} (R_i \cap M_j) = M_i,$$

i.e., v^* has center M_i in R_i for each i . Again by Proposition 3 and Theorem 1, it follows that the R_i -dimension of v^* is zero for each i . Finally, if v is any other valuation of K with center M_i in R_i for each i then $R_v \supset \bigcup_{i=0}^{\infty} R_i = R_v$, and $M_v \cap R_v = \bigcup_{i=0}^{\infty} (M_v \cap R_i) = \bigcup_{i=0}^{\infty} M_i = M_v$, and hence $v = v^*$.

LEMMA 13. Let (R, M) be a regular two dimensional local domain with quotient field K , let P be a minimal prime ideal in R , let $\bar{R} = R/P$ and $\bar{M} = M/P$. Let $\bar{K}$ be the quotient field of $\bar{R}$ and let T be the integral closure of $\bar{R}$ in $\bar{K}$. Assume that R is the quotient ring of a point either on an algebraic surface or on an absolute surface (i.e., K is a one dimensional algebraic function field over Q and R is the quotient ring of a finitely generated domain over Z with respect to a maximal ideal). Then T is a finite R -module.

Proof. The case of algebraic surfaces is well known (see p. 511 of [10]) and so we may assume that R is the quotient ring of a point on an absolute surface. Then we can find $z_1, z_2, \dots, z_s$ in K such that K is the quotient field of $R^* = Z[z_1, z_2, \dots, z_s]$ and such that $R = R^*_{M^*}$, where M^* is a maximal ideal in R^* . Let $P^* = R^* \cap P$, $\bar{R}^* = R^*/P^*$, $\bar{M}^* = M^*/P^*$, and $\bar{z}_i =$ the P^* -residue class of z_i . Then $\bar{R} = \bar{R}^*_{\bar{M}^*}$. If $Z \cap P^* \neq (0)$ then $Z \cap P^* = pZ$ where p is a prime number and $\bar{R}^* = (Z/pZ)[\bar{z}_1, \bar{z}_2, \dots, \bar{z}_s]$; hence we are again in the algebro-geometric case. Now assume that $Z \cap P^* = (0)$. Let $S = R^*_{P^*}$, $N = P^*S$ and $\bar{S} = S/N$. Then $\bar{S} = \bar{R}$. Also $Q \subset S$ and hence $S = S^*_{N^*}$, where $S^* = Q[z_1, z_2, \dots, z_s]$ and $N^* = S^* \cap N$. Since S^* is of transcendence degree one over Q , it follows that $\bar{K}$ is a finite algebraic extension of Q . Thus it is enough to prove the following assertion: Let L be a finite algebraic extension of Q , let A be a proper subdomain of L having L as quotient field, and let B be the integral closure of A in L . Then B is a finite A -module.

First let us recall that if D is a commutative ring with 1, and W the set of all maximal ideal in D then $D = \bigcap_{w \in W} D_w$. For if $d \in \bigcap_{w \in W} D_w$ then $d = e_w/f_w$ with $e_w, f_w \in D$ and $f_w \notin w$. Now since the ideal E generated by all the f_w

is not contained in any member of W , we must have $E=D$. Therefore we can write $1=f_{w_1}g_1+f_{w_2}g_2+\cdots+f_{w_n}g_n$ with g_i in D and hence $d=e_{w_1}g_1+e_{w_2}g_2+\cdots+e_{w_n}g_n\in D$. Therefore $D=\bigcap_{w\in W}D_w$. To prove the

italicized assertion, let Y be the integral closure of Z in L . Then Y is a finite Z -module; let $y_1, y_2, \cdots, y_n$ be a module basis of Y over Z . Let $C=A[y_1, y_2, \cdots, y_n]$. Then it is clear that $y_1, y_2, \cdots, y_n$ is a module basis of C over A . Let M be a maximum ideal in C , and let $m=Y\cap M$. Then $Y_m\subset C_m$. If $m=(0)$ then $Y_m=L$ and if $m\neq(0)$ then Y_m is the valuation ring of real discrete valuation of L , i.e., Y_m is a maximal subring of L . Therefore $Y_m=C_m$. Hence by the above observation, C is an intersection of valuation rings. Therefore C is integrally closed in L . Since $C\subset B$, we must have $C=B$, i.e., B is a finite A -module.

In the following proposition we shall prove that the singularities of a curve lying on a nonsingular *absolute surface* can be resolved by quadratic transformations applied to the surface. We shall base our demonstration on Zariski's proof of the algebro-geometric case; see Theorem 4 of [11].

PROPOSITION 5. *Let (R, M) be a two dimensional regular quotient ring of a point on an algebraic or absolute surface and let K be the quotient field of R . Let P be a minimal prime ideal in R and let w be the valuation of K with $R_w=R_P$. Let v be a valuation of K composite with w and having center M in R . Let (R_n, M_n) be the n -th quadratic transform of R along v and let $P_i=R_i\cap M_v$. Then P_i is a minimal prime ideal in R_i for $i=1, 2, \cdots$; and there exists an integer n such that for any $i\geq n$ we can choose a basis (x_i, y_i) of M_i for which $x_iR_i=P_i$, $v(y_i)=(0, 1)$, and $v(x_i)=(1, a)$ where a is some integer.*

Proof. Since $M_i\cap R=M$ and $P_i\cap R=R\cap M_w=P$ and since R_i is two dimensional, P_i must be a minimal prime ideal in R_i . Now let (x, y) be a basis of M , and suppose for instance that $v(x)\leq v(y)$. Then $w(x)\leq w(y)$. Therefore $x\notin P$; for otherwise we would have $w(x)>0$ and hence $w(y)>0$, i.e., $x\in R\cap M_w=P$ and $y\in R\cap M_w=P$, and hence $M=P$, which is a contradiction. Let $R^*=R[y/x]$, $M^*=R^*\cap M_v$, $P^*=R^*\cap M_w$. Let $\bar{R}=R/P$, $\bar{M}=M/P$, $\bar{R}^*=R^*/P^*$, $\bar{M}^*=M^*/P^*$, $\bar{R}_i=R_i/P_i$, $\bar{M}_i=M_i/P_i$, $\bar{R}=R_w/M_w$, $\bar{v}$ —the valuation of $\bar{R}$ induced by v , and $\bar{x}, \bar{y}$ the residue classes modulo M_w respectively of x, y . Then $(\bar{R}_i, \bar{M}_i)$ is a one dimensional local domain with quotient field $\bar{K}$, the real discrete valuation $\bar{v}$ of $\bar{K}$ has center M_i in $\bar{R}_i$, and $\bar{R}=\bar{R}_0\subset\bar{R}_1\subset\bar{R}_2\subset\cdots$. Now $R_1=R^*_{M^*}$, $M_1=M_0R_1$ and $0\neq\bar{x}\in\bar{R}$. Therefore $(\bar{x}, \bar{y})\bar{R}=\bar{M}$, $\bar{R}^*=\bar{R}[\bar{y}/\bar{x}]$, $\bar{M}^*=\bar{R}^*\cap\bar{M}_0$ and

$\bar{R}_1 = \bar{R}^* \bar{M}^*$. Hence $\bar{M} \bar{R}^* = \bar{x} \bar{R}^*$ and $\bar{M} \bar{R}_1 = \bar{x}_1 \bar{R}_1$. Similarly $\bar{M}_i \bar{R}_{i+1} = z_{i+1} \bar{R}_{i+1}$ with $z_{i+1} \in \bar{R}_{i+1}$ for $i = 0, 1, 2, \dots$.

We shall now show that $\bigcup_{i=0}^{\infty} \bar{R}_i = R_v$. Given c in R_v we can write $c = a/b$ with $0 \neq b, a \in \bar{R}$. If $b \notin \bar{M}$ then $c \in \bar{R}$. Suppose that $b \in \bar{M}$. Since $\bar{v}(a) \geq \bar{v}(b)$ we must have $a \in \bar{M}$. Hence $b = b_1 z_1, a = a_1 z_1$ with a_1, b_1 in $\bar{R}_1$. Since $c = a_1/b_1$, again either $c \in \bar{R}_1$ or $a_1, b_1 \in \bar{M}_1$. Suppose that $c \notin \bar{R}_1$. Then $a_1 = a_2 z_2, b_1 = b_2 z_2$ with $a_2, b_2 \in \bar{R}_2$. Similarly if $c \notin \bar{R}_{n-1}$ then $a = a_n z_1 z_2 \cdots z_n$ and $b = b_n z_1 z_2 \cdots z_n$ with $a_i, b_i \in \bar{R}_i$. Since $\bar{M}_{n-1} = \bar{R}_{n-1} \cap \bar{M}_v$, we must have $\bar{v}(z_n) > 0$, and hence that $\bar{v}(b) \geq n$ where we take Z as the value group of the real discrete valuation $\bar{v}$. Since $\bar{v}$ is real discrete and since $b \neq 0$, for some n we must have $c \in \bar{R}_n$. Therefore $\bigcup_{i=0}^{\infty} \bar{R}_i = R_v$. Let S_i be the integral closure of $\bar{R}_i$ in $\bar{K}$. Then S_i is a Dedekind domain with a finite number of prime ideals (§39 of [5]), and since the quotient ring of S_i with respect to any prime ideal is a real discrete valuation ring, we must have $S_i = \bigcap_{u \in W_i} R_u$ where W_i is a finite set of real discrete valuations of $\bar{K}$. It is clear that the valuations in W_i are exactly the valuations of $\bar{K}$ having center $\bar{M}_i$ in $\bar{R}_i$. Therefore $W_i \supset W_{i+1}$. Let $u_1, u_2, \dots, u_h$ be the valuations in W_0 different from $\bar{v}$. Since R_v is a maximal subring of $\bar{K}$, we can find $a_i \in R_v$ such that $a_i \notin R_{v_i}$. Since $\bigcup_{j=0}^{\infty} \bar{R}_j = \bar{R}_v, a_i \in \bar{R}_{m_i}$ for some integer m_i . Let $m = \max(m_1, m_2, \dots, m_h)$. Then $a_i \notin \bar{R}_m$ for $i = 1, 2, \dots, h$. Therefore $W_m = \{\bar{v}\}$, i.e., R_v is the integral closure of $\bar{R}_m$. By Lemma 13, R_v is a finite module over $\bar{R}_m$. Hence by the Hilbert basis theorem, we can find n such that $\bar{R}_i = \bar{R}_v$ whenever $i \geq n$; from now on we shall assume that $i \geq n$. Fix $\bar{y}_i$ in $\bar{R}_i$ with $\bar{v}(\bar{y}_i) = 1$. Let y_i be an element of R_i belonging to the residue class $\bar{y}_i$. Then $v(y_i) = (0, 1)$. By Lemma 9, P_i is a principal ideal; let $P_i = x_i R_i$. Since $(R_i)_{P_i} = R_v, v(x_i) = (1, a)$. Finally given z in M_i let $\bar{z}$ be the residue class of z modulo P_i . Then $\bar{z} \in \bar{M}_i$ and hence $\bar{z} = \bar{y}_i \bar{t}$ with $\bar{t} \in \bar{R}_i$. Fix t in R_i belonging to the residue class $\bar{t}$. Then $z - y_i t \in P_i$, i.e., $z \in (x_i, y_i) R_i$. Therefore $(x_i, y_i) R_i = M_i$.

Remark 3. Observe that Lemma 13, which is crucial in the proof of Proposition 5, breaks down for certain abstract local rings (regular of dimension two) as can be shown by using an example due to F. K. Schmidt; see page 24 of Zariski's paper [12]. To indicate this we shall use Zariski's notation. We have then two independent variables x and t over a certain field k of characteristic $p \neq 0$, $\Sigma' = k(x, t)$, σ' is the valuation ring of a certain real discrete valuation v' of Σ'/k , $\Sigma = \Sigma'(\tau)$ with $\tau = t^{1/p}$ and σ is the

one dimensional local domain $\mathfrak{o}'[\tau]$ with quotient field Σ . Let T be the integral closure of $\mathfrak{o}'$ in Σ . Since $\mathfrak{o}$ is integral over $\mathfrak{o}'$, T is also the integral closure of $\mathfrak{o}$ in Σ . Let $R = \mathfrak{o}'[X]$ and $P = (X^2 - t)R$. Then it is easily verified that R is a two dimensional regular local domain, P is a minimal prime ideal in R , and $X \rightarrow \tau$ is an $\mathfrak{o}'$ -homomorphism of R onto $\mathfrak{o}$ with kernel P , i. e., $\mathfrak{o}$ is isomorphic to R/P . Since T is not a finite $\mathfrak{o}'$ -module (see part c on page 447 of [9]) and since $\mathfrak{o}$ is a finite $\mathfrak{o}'$ -module, T cannot be a finite $\mathfrak{o}$ -module. Thus Lemma 13 is not applicable to the minimal prime ideal P of the two dimensional regular local domain R .

Now let K be the quotient field of R , let M be the maximal ideal in R , let w be the valuation of K with $R_w = R_P$, and let v be a valuation of K composed with w and having center M in R . Suppose, if possible, that Proposition 5 were true for the regular local domain (R, M) . Then, in the notation of Proposition 5, we would have that R_n/P_n is a valuation ring. From the considerations made in the proof of Proposition 5, it follows that this would imply that $\mathfrak{o}$ can be transformed into a regular local ring by applying to $\mathfrak{o}$, n successive quadratic transformations in the sense described on page 24 of [12], but this is impossible as is proved on pages 24-25 of [12]. Therefore, Proposition 5 is not applicable to the valuation v having center M in the regular two dimensional local domain R . Since Proposition 5 is essentially based on Lemma 13,⁵ this again shows, as we have directly proved above, that Lemma 13 is not valid for arbitrary two dimensional regular local domains.

THEOREM 2.⁶ Let (R, M) be a two dimensional regular local domain with quotient field K . Let v be a valuation of K having center M in R and R -dimension zero. Let f be a given nonzero element of R . Then there exists a quadratic transform (R^*, M^*) of R along v and a basis (x^*, y^*) of M^* such that $f = x^{*a}y^{*b}d$ where a and b are nonnegative integers, d is a unit in R^* , and where the following conditions are satisfied: (A) If v is real of rational rank one, then $b = 0$. (B) If v is real of rational rank two then either $b = 0$ or $v(x^*)$ and $v(y^*)$ form an integral basis for the value group of v . (C) If v is of rank two and if R is the quotient ring of a point either on an algebraic surface or on an absolute surface, then $v(x^*) = (1, h)$ and $v(y^*) = (0, 1)$ where we are writing the v -values of elements of K as lexicographically ordered pairs of integers and where h is some integer.

⁵ I. e., the only fact about valuations centered in local domains which is used in the proof of Proposition 5 and in which the local domains are qualified to be either algebraic or absolute is Lemma 13.

⁶ See Proposition 3 of [2].

Proof. First assume that v is real. Let $R_0 = R$ and $M_0 = M$. Let (x_0, y_0) be a basis of M_0 and let (R_i, M_i) be the i -th quadratic transform of R_0 along v . We shall define elements (x_i, y_i) of R_i by induction on i . Let then $i = m > 0$ and assume that we have defined x_i, y_i for $i = 1, 2, \dots, m-1$. Suppose first that $v(y_{i-1}) \geq v(x_{i-1})$. Let $S_i = R_{i-1}[y_{i-1}/x_{i-1}]$ and $N_i = M_{i-1}S_i$. Let $P_i = M_v \cap S_i$. Let z_i be the residue class modulo N_i containing y_{i-1}/x_{i-1} . Then by Lemma 3 of [1], $S_i/N_i = (R_{i-1}/M_{i-1})[z_i]$ and z_i is transcendental over R_{i-1}/M_{i-1} . Since P_i is a maximal ideal in S_i containing N_i , P_i/N_i must be a maximal ideal in S_i/N_i and hence $P_i/N_i = g_i(z_i)(S_i/N_i)$ where $g_i(X)$ is a monic irreducible polynomial in $(R_{i-1}/M_{i-1})[X]$. Let $G_i(X)$ be a monic polynomial in $R_{i-1}[X]$ which when reduced modulo M_{i-1} gives $g_i(X)$. We set $x_i = x_{i-1}$ and $y_i = G_i(y_{i-1}/x_{i-1})$. Secondly, if $v(y_{i-1}) < v(x_{i-1})$ then we set $x_i = y_{i-1}$ and $y_i = x_{i-1}/y_{i-1}$. Then by Corollary 1 of [1] $(x_i, y_i)R_i = M_i$.

Let $f_0 = f$ and define by induction $f_i \in R_i$ by the equation $f_{i-1} = x_i^{u_i} f_i$ where u_i is a nonnegative integer and where f_i is an element in R_i prime to x_i (by Lemma 9, R_i is a unique factorization domain). Let $f_{i,1}, f_{i,2}, \dots, f_{i,\lambda_i}$ be the irreducible factors of f_i in R_i and let $w_{i,j}$ be the valuation of K whose valuation ring is the quotient ring of R_i with respect to $f_{i,j}R_i$. Let W_i be the set of valuations u of K such that u has center M_i in R_i and u is composed with $w_{i,j}$ for some j . By Lemma 8, W_i is a finite set. For a given u in W_i , let $P_i = M_u \cap R_{i-1}$. Since u is nontrivial, $P_i \neq (0)$. Since $x_i \notin M_u \cap R_i$ and since $x_i R_i = M_{i-1} R_i$, $P_i \neq M_{i-1}$. Therefore P_i is a minimal prime ideal in R_{i-1} . Since R_{i-1} is a unique factorization domain and since $f_{i-1} \in P_i$, we must have $P_i = f_{i-1,j} R_{i-1}$ for some j , i.e., $u \in W_{i-1}$. Thus $W_i \subset W_{i-1}$. Since by Lemma 12, $\bigcup_{i=1}^{\infty} R_i = R_v$, since v is real, and since no element of W_i is real, it follows that $\bigcap_{i=1}^{\infty} W_i = \emptyset$. Since W_i is finite, $W_m = \emptyset$ for some m , i.e., f_m

is a unit in R_m . Hence we have $f = x_m^A y_m^B D$ where A and B are nonnegative integer and D is a unit in R_m . If either v is of rational rank one or if v is of rational rank two and $v(x_m)$ and $v(y_m)$ are rationally dependent, then the proof can be completed by the argument of the last part of the proof of Proposition 3 of [2]. Now suppose that v is of rational rank two and that $v(x_m)$ and $v(y_m)$ are rationally independent. We may then take R^* to be R_m and $x^* = x_m, y^* = y_m$. It then remains to be shown that $v(x^*)$ and $v(y^*)$ form an integral basis for the value group of v . Let $v(x^*) = p$ and $v(y^*) = q$; and suppose for instance that $p < q$. Fix a representative system k in R^* of R^*/M^* (k is not in general a field; note that we take zero as the representative of M^*/M^*). Let z be an arbitrary nonzero element of R^* and let $v(z) = r$. Fix an integer n so that $r < np$. Then we can find

a polynomial $H(X, Y) = \sum_{i+j \leq n} H_{ij} X^i Y^j$ of degree at most n with coefficients H_{ij} in k such that $z^* = z - H(x^*, y^*) \in M^{*n}$ (see § 5 of [6]). Since $v(u) > r$ for any $u \in M^{*n}$, we must have that $v(z^*) > r$ and hence that $r = v(z) = v(H(x^*, y^*))$. Since p and q are rationally independent and since $v(H_{ij}) = 0$ whenever $H_{ij} \neq 0$, we can find $H_{st} \neq 0$ such that $v(H_{st} x^{*s} y^{*t}) < v(H_{ij} x^{*i} y^{*j})$ whenever $H_{ij} \neq 0$ and whenever either $i \neq s$ or $j \neq t$. Therefore $r = v(z) = v(H(x^*, y^*)) = v(H_{st} x^{*s} y^{*t}) = sp + tq$. Therefore for any nonzero element z_1 of K , we must have $v(z_1) = s_1 p + t_1 q$ where s_1 and t_1 are integers. Thus we have shown that $\{p, q\}$ is an integral basis of the value group of v . This completes the proof of (A) and (B).

To prove (C), assume that v is of rank two, that R is the quotient ring of a point either on an algebraic or on an absolute surface and that we are writing the values of elements of K as lexicographically ordered pairs of integers. By Proposition 5, we can find a quadratic transform $(\bar{R}, \bar{M})$ of R along v and a basis $\bar{x}, \bar{y}$ of $\bar{M}$ such that $v(\bar{x}) = (1, p)$ and $v(\bar{y}) = (0, 1)$ where p is some integer. Let $(\bar{R}_i, \bar{M}_i)$ be the i -th quadratic transform of $\bar{R}$ along v . Let $\bar{x}_i = \bar{x}/\bar{y}_i$ and $\bar{y}_i = \bar{y}$. Since $v(\bar{x}) > iv(\bar{y})$ for any integer i , it follows that $(\bar{x}_i, \bar{y}_i) \bar{R}_i = \bar{M}_i$. Let $f = \bar{x}^a g_0$ where a is a nonnegative integer and where g_0 is an element of $\bar{R}$ prime to $\bar{x}$. Then $v(g_0) = (0, t)$ where t is some nonnegative integer. Define g_i by induction by the equation $g_{i-1} = \bar{y}_i^{e_i} g_i$ where e_i is a nonnegative integer and g_i is an element of $\bar{R}_i$ prime to $\bar{y}_i$. Since $M_{i-1} \bar{R}_i = \bar{y}_i \bar{R}_i$, we have that $e_i > 0$ whenever g_{i-1} is a nonunit in $\bar{R}_{i-1}$ (i.e., whenever $g_{i-1} \in \bar{M}_{i-1}$). Therefore, if g_{i-1} is a nonunit in $\bar{R}_{i-1}$ for $i = 1, 2, \dots, n$, then $v(g_0) \geq (0, n)$. Hence for some integer $m \leq t$, g_m must be a unit in $\bar{R}_m$. Hence $f = x^{*a} y^{*b} d$ where b is a nonnegative integer, d is a unit in $R^* = \bar{R}_m$, $x^* = \bar{x}_m$, and $y^* = \bar{y}_m$.

We shall now prove a generalization to abstract two dimensional regular local domains of Zariski's theorem on the factorization of birational transformation between nonsingular algebraic surfaces into local quadratic transformations; see the lemma on page 538 of [11].

THEOREM 3. *Let (R, M) and (R', M') be regular two dimensional local domains with a common quotient field K such that (R', M') has center M in R . Then R' is a quadratic transform of R .*

We shall precede the proof proper of this theorem by three preparatory lemmas.

LEMMA 14. *Let A be a normal domain with quotient field K such that A contains a unique maximal ideal P and let $0 \neq x \in K$ such that $x \notin A$ and*

$(1/x) \notin A$. Let $B = A[x]$ and let $M = PB$. Then: (1) there exists a valuation v of K such that $R_v \supset B$, $M_v \cap B = M$; (2) $M \cap A = P$; (3) the residue class $\bar{x}$ modulo M containing x is transcendental over A/P , i.e., R_v/M_v contains the polynomial ring $(R/M)[\bar{x}]$ and hence R_v/M_v is of positive transcendence degree over R/M .

Proof. Follows by the considerations made by Zariski on pages 26-27 of [14].

LEMMA 15. Let (R, M) be an n dimensional regular local domain with quotient field K and $n > 1$. Let $\{x_1, x_2, \dots, x_n\}$ be a minimal basis of M . Let $A = R[x_2/x_1, x_3/x_1, \dots, x_n/x_1]$ and $P = MA$. Then: (1) $P = x_1A$; (2) P is a minimal prime ideal of A ; (3) A_P is the valuation ring of a real discrete valuation w of K ; (4) $A/P = (R/M)[\bar{y}_2, \bar{y}_3, \dots, \bar{y}_n]$ where $\bar{y}_i =$ the residue class modulo P containing x_i/x_1 , and $\bar{y}_2, \bar{y}_3, \dots, \bar{y}_n$ are algebraically independent over R/M ; (5) $R_w/M_w = (R/M)(\bar{y}_2, \bar{y}_3, \dots, \bar{y}_n)$; and (6) w is the " M -adic divisor of R ," i.e., w is completely defined by setting $w(a/b) = (\text{leading degree of } a) - (\text{leading degree of } b)$ where a and b are any two nonzero elements of R , and w is of R -dimension $(n-1)$.

Proof. (1) is obvious, (2) and (4) are proved in Lemma 3 of [1]. Now let N be a maximal ideal in A containing P , let $R^* = A_N$ and $M^* = NR^*$. Then by Corollary 1 of Lemma 3 of [1], R^* is a regular local domain and $P^* = y_1R^*$ is a minimal prime ideal of R^* . Hence $R^*_{P^*} = R_w$ where w is a real discrete valuation of K . Now (3) follows since $A_P = R^*_{P^*}$, and (5) follows from (4) and the fact that $A_P/(PA_P)$ is the quotient field of A/P . Now let a be an arbitrary nonzero element of R and let d be the leading degree of a . Then $a = f(x_1, x_2, \dots, x_n)$ where $f(X_1, X_2, \dots, X_n)$ is a form of degree d with coefficients in R not all in M ; and hence $a = x_1^d g$ where $g = f(1, x_2/x_1, x_3/x_1, \dots, x_n/x_1) \notin P$ since $\bar{y}_2, \bar{y}_3, \dots, \bar{y}_n$ are algebraically independent over R/M ; i.e., $w(a) = d$. Finally, it follows by (5) that w is of R -dimension $(n-1)$ and this proves (6).

LEMMA 16. Let (R, M) and (R', M') be as in Theorem 3 and assume that $R \neq R'$. Then there exists a nonunit z in R' such that $MR' \subset zR'$.

Proof. Suppose, if possible, that MR' is primary for M' . Let $(\bar{R}, \bar{M})$ and $(\bar{R}', \bar{M}')$ be the completions of R and R' respectively. Let w be the M' -adic divisor of R' . Then w has center M in R and R -dimension of $w \geq R'$ -dimension of $w = 1$. Therefore by Theorem 1, w is a prime divisor also for R . Hence by Theorem 2 of [13], (R, M) and (R', M') are both subspaces

of (R_w, M_w) and hence R is a subspace of R' . Therefore we may canonically assume that $\bar{R}$ is a subring and a subspace of $\bar{R}'$. By Theorem 1, R_w/M_w is finitely generated over R/M and hence R'/M' must be a finite algebraic extension of R/M . Since $\bar{R}'/\bar{M}' = R'/M'$ and $\bar{R}/\bar{M} = R/M$, we have that $\bar{R}/\bar{M}$ is a finite algebraic extension of R/M . Also MR' is primary for M' implies that $\bar{M}'\bar{R}'$ is primary for $\bar{M}'$. Therefore by Theorem 8 of [3], $\bar{R}'$ is integral over $\bar{R}$. Let E and E' be the quotient fields of $\bar{R}$ and $\bar{R}'$ respectively. Let A be the set of valuations v of K having center $\bar{M}$ in $\bar{R}$ and let B be the set of valuations u of K for which $R_u \supset R$. Let v be a member of A . Then by Lemma 13 of [2], v has an extension $\bar{v}$ to E having center $\bar{M}$ in $\bar{R}$. Since $\bar{R}'$ is integral over $\bar{R}$, we must have $R_{\bar{v}} \supset \bar{R}'$ for some extension $\bar{v}'$ of $\bar{v}$ to E' . Since $R_v = R_{\bar{v}} \cap K$ and since $R' = \bar{R}' \cap K$ (Lemma 2 of [2]), we must have $R' \subset R_v$. Therefore $R' \subset \bigcap_{v \in A} R_v$. Now let u be a member of B for which $P = M_u \cap R \neq M$. Let $\bar{K} = R_u/M_u$, $\bar{R} = R/P$ and $\bar{M} = M/P$. Let $\bar{u}$ be a valuation of $\bar{K}$ having center $\bar{M}$ in $\bar{R}$ and let u^* be the valuation of K which is composed of u and $\bar{u}$. Then $R_{u^*} \supset R_u$ and u^* has center M in R . Therefore $R' \subset \bigcap_{u \in B} R_u = R$, i. e., $R' = R$. Thus our assumption that MR' is primary for M' is absurd. Since R' is two dimensional, $MR' \subset p$ where p is a minimal prime ideal in R' and by Lemma 9, $p = zR'$ with $z \in R'$, i. e., $MR' \subset zR'$.

Proof of Theorem 3. If $R = R'$ then there is nothing to prove. So assume that $R < R'$. By Lemma 9, there exists a discrete rank two valuation v of K having center M' in R' . By Theorem 1, R -dimension of $v = 0 = R'$ -dimension of v . Let $\{x, y\}$ be a minimal basis of M , and suppose for instance that $v(x) \leq v(y)$. Let $t = y/x$. If $1/t \in R'$ then $t \in R_v$ and $1/t \in R_v$ and hence $1/t \notin M_v$, i. e., $1/t \notin M'$ and hence $t \in R'$. Now suppose, if possible, that $t \notin R'$. Then $1/t \notin R'$. Let $A' = R'[t]$ and $P' = M'A'$. By Lemma 14, there exists a valuation w of K with center P in A , and

$$R_w/M_w \supset A'/P' = (R'/M')[\bar{t}]$$

where $\bar{t}$ is the residue class modulo P' containing t and $\bar{t}$ is transcendental over R'/M' . Let $A = R[t]$, $P = MA$ and $P^* = P' \cap A$. Then $P^* \cap R = M$ and hence t is *a fortiori* transcendental over R modulo M . Hence $P^* = MA = P = xA$. Therefore $A_P \supset R_w$ and hence by Lemma 15, w must be the M -adic divisor of R . Since $w(x) = 1$ and since $w(M') > 0$, we have that $x \in M'$ and $x \notin (M')^2$, i. e., that x is an irreducible nonunit in R' . By Lemma 16, $x = az$ and $y = bz$ where a and b are in R' and z is a nonunit in R' . Since by Lemma 9, R' is a unique factorization domain, we must have $y = cz$

with c in R' , i. e., $t = c \in R'$. Thus our assumption that $t \notin R'$ is absurd. Let (R_i, M_i) be the i -th quadratic transform of R along v . Then $R_1 \subset R'$ and R' has center M_1 in R_1 . Similarly, if $R_1 \subset R'$ then $R_2 \subset R'$ and R' has center M_2 in R_2 and so on. Suppose, if possible, that $R_i \subset R'$ for all i . Then by Lemma 12, $R_v \subset R'$ and hence $R_v = R'$. Since v is rank two, R_v , i. e., R' is nonnoetherian which is absurd. Therefore, for some integer n , we must have $R_n = R'$.

LEMMA 17. *Lemma 12 remains true if we replace the assumption that R_{i+1} is a quadratic transform of R_i for $i = 0, 1, 2, \dots$, by the weaker assumption that $M_{i+1} \cap R_i = M_i$ for $i = 0, 1, 2, \dots$.*

Proof. Follows by Lemma 12 and Theorem 3.

Appendix.*

That an ordered abelian group G of finite rank can be embedded (as an ordered subgroup) in the lexicographically ordered direct sum $R^{(\rho)}$ of ρ copies of the additive group R of real numbers can be proved as follows: First we prove that if G is *rationaly complete* (i. e., with any element g of G and an arbitrary nonzero integer n , G always contains an element g^* such that $ng^* = g$) then we have the stronger result that G itself can be expressed as a lexicographically ordered direct sum of ordered subgroups of rank one, i. e., if $0 = G_0 < G_1 < \dots < G_\rho = G$ is the sequence of isolated subgroups of G then G contains ordered subgroups $H_1, H_2, \dots, H_\rho$ of rank one such that $G_{\rho-i} = H_{i+1} \oplus H_{i+2} \oplus \dots \oplus H_\rho$ where the sum is lexicographically ordered direct (i. e., every element g of $G_{\rho-i}$ has a unique expression $g = h_{i+1} + h_{i+2} + \dots + h_\rho$ with h_j in H_j , if $g^* = h_{i+1}^* + h_{i+2}^* + \dots + h_\rho^*$ with h_j^* in H_j is any other element of $G_{\rho-i}$ different from g and if $h_{i+j} = h_{i+j}^*$ for $j = 1, 2, \dots, t-1$ and $h_{i+t} \neq h_{i+t}^*$ then $g < g^*$ or $g > g^*$ according as $h_{i+t} < h_{i+t}^*$ or $h_{i+t} > h_{i+t}^*$). For $\rho = 1$ this is obvious and to apply induction, assume that $\rho > 1$ and that the assertion is true for $\rho - 1$. Let H be the maximal isolated subgroup of G . Then H is of rank $\rho - 1$, G/H is of rank one, and H and G/H are both rationally complete. By the induction hypothesis, $H = H_2 \oplus H_3 \oplus \dots \oplus H_\rho$ where the sum is lexicographically ordered direct. Let B be a rationally independent rational basis of G/H and for each b in B fix β in G contained in the residue class b . Let H_1 be the set of all elements of G which depend rationally on the β 's. Then it can be easily verified that

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H_1 is order isomorphic to G/H and that $G = H_1 \oplus H = H_1 \oplus H_2 \oplus \cdots \oplus H_p$ where the direct sums are lexicographically ordered.

In the general case when G is not necessarily rationally complete, it is enough to observe that G can always be embedded in a rational completion i.e. in an abelian group G^* which is rationally complete and for any element g^* of G we have $ng^* \in G$ for some nonzero integer n . The ordering of G can be uniquely extended to G^* and then G^* is again of rank ρ . As shown above, we can find in G^* subgroups $H_1, H_2, \dots, H_p$ of rank one such that $G^* = H_1 \oplus H_2 \oplus \cdots \oplus H_p$ where the direct sum is lexicographically ordered. Now each H_i being archimedean is order isomorphic to a subgroup of R and hence G^* is order isomorphic to a subgroup of $R^{(\rho)}$. Therefore G is order isomorphic to a subgroup of $R^{(\rho)}$. Now we shall show that if G is not rationally complete then it need not be order isomorphic to a lexicographically ordered direct sum of rank one groups. To show this, let G denote the torsionfree abelian group studied by L. Pontrijagin in Example 2 of Appendix 1 of his paper: The theory of topological commutative groups; *Annals of Mathematics*, Vol. 35 (1934), pp. 361-388. Then G is of rational rank two and it is not expressible as a direct sum of rational rank one subgroups. Let (u, v) be a rational basis of G . Then each element g of G has a unique expression $g = au + bv$ where a and b are rational numbers. Let $g^* = a^*u + b^*v$ be any other element of G where a^* and b^* are rational numbers. We set $g > g^*$ either if $a > a^*$ or if $a = a^*$ and $b > b^*$. This turns G into an ordered abelian group and we have $\rho(G) = 2$. Suppose if possible that G is the lexicographically ordered direct sum of two ordered rank one subgroups H_1 and H_2 . Then we have that $r(H_1) = r(H_2) = 1$ and hence that G is the direct (group theoretic) sum of two rational rank one subgroups; this is absurd.

We observe that if in the above proof we replace simple induction by transfinite induction then we obtain the following more general result: Let G be an ordered abelian group and let $S(G)$ be the family of isolated subgroups of G simply ordered by inclusion. Assume that $S(G)$ is well ordered. Let G^* be a rational completion of G and extend the ordering of G to G^* (uniquely). Then G^* is order isomorphic to a lexicographically ordered direct sum of a well ordered family T [which is order isomorphic to $S(G)$] of archimedean groups.

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